

Гадиров Наиль

задание 20_1

1) создаем модуль, мейк

```
#include <linux/module.h>
#include <linux/kernel.h>

int init_module(void)
{
    pr_info("My test module loaded!!!!\n");

    return 0;
}

void cleanup_module(void)
{
    pr_info("My test module unloaded!!!!\n");
}

MODULE_LICENSE("GPL");
```

```
obj-m += test1.o
#ccflags-y += -g -DDEBUG

all:
    make -C /lib/modules/6.17.0-35-generic/build M=$(PWD) modules

clean:
    make -C /lib/modules/6.17.0-35-generic/build M=$(PWD) clean
```

2) собираем

```
gadirov@nail-gad:~/Desktop/HW_20$ make
make -C /lib/modules/6.17.0-35-generic/build M=/home/gadirov/Desktop/HW_20 modules
make[1]: вход в каталог «/usr/src/linux-headers-6.17.0-35-generic»
make[2]: вход в каталог «/home/gadirov/Desktop/HW_20»
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
  You are using:          gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
  CC [M]  test1.o
  MODPOST Module.symvers
WARNING: modpost: missing MODULE_DESCRIPTION() in test1.o
  CC [M]  test1.mod.o
  CC [M]  .module-common.o
  LD [M]  test1.ko
  BTF [M] test1.ko
Skipping BTF generation for test1.ko due to unavailability of vmlinux
make[2]: выход из каталога «/home/gadirov/Desktop/HW_20»
make[1]: выход из каталога «/usr/src/linux-headers-6.17.0-35-generic»
gadirov@nail-gad:~/Desktop/HW_20$
```

3) insmod и dmesg и lsmod и rmmod

```
gadirov@nail-gad:~/Desktop/HW_20$ sudo insmod test1.ko
gadirov@nail-gad:~/Desktop/HW_20$ sudo dmesg | tail -n10
[60596.797203] audit: type=1400 audit(1781786046.851:779): apparmor="ALLOWED" operation="open" class="file" profile="libreoffice-soffice" name="/sys/fs/cgroup/cgroup.controllers" pid=405476 comm="soffice.bin" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[60596.797220] audit: type=1400 audit(1781786046.851:780): apparmor="ALLOWED" operation="file_perm" class="file" profile="libreoffice-soffice" name="/sys/fs/cgroup/cgroup.controllers" pid=405476 comm="soffice.bin" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
[60596.797227] audit: type=1400 audit(1781786046.851:781): apparmor="ALLOWED" operation="open" class="file" profile="libreoffice-soffice" name="/proc/405476/cgroup" pid=405476 comm="soffice.bin" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[60596.797232] audit: type=1400 audit(1781786046.851:782): apparmor="ALLOWED" operation="file_perm" class="file" profile="libreoffice-soffice" name="/proc/405476/cgroup" pid=405476 comm="soffice.bin" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[60596.797237] audit: type=1400 audit(1781786046.851:783): apparmor="ALLOWED" operation="file_perm" class="file" profile="libreoffice-soffice" name="/proc/405476/cgroup" pid=405476 comm="soffice.bin" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[60596.797242] audit: type=1400 audit(1781786046.851:784): apparmor="ALLOWED" operation="open" class="file" profile="libreoffice-soffice" name="/proc/405476/mountinfo" pid=405476 comm="soffice.bin" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[60596.797246] audit: type=1400 audit(1781786046.851:785): apparmor="ALLOWED" operation="file_perm" class="file" profile="libreoffice-soffice" name="/proc/405476/mountinfo" pid=405476 comm="soffice.bin" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[60596.797250] audit: type=1400 audit(1781786046.851:786): apparmor="ALLOWED" operation="file_perm" class="file" profile="libreoffice-soffice" name="/proc/405476/mountinfo" pid=405476 comm="soffice.bin" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[60596.797252] audit: type=1400 audit(1781786046.851:787): apparmor="ALLOWED" operation="file_perm" class="file" profile="libreoffice-soffice" name="/proc/405476/mountinfo" pid=405476 comm="soffice.bin" requested_mask="r" denied_mask="r" fsuid=1000 ouid=1000
[61608.477893] My test module loaded!!!!
gadirov@nail-gad:~/Desktop/HW_20$
```

```
gadirov@nail-gad:~/Desktop/HW_20$ lsm
lsmem  lsmod
gadirov@nail-gad:~/Desktop/HW_20$ lsmod
Module                Size  Used by
test1                  12288  0
exfat                  110592  1
nfnetlink_queue       28672  1
nft_queue              12288  2
xt_conntrack           12288  3
xt_MASQUERADE          16384  3
```

```
gadirov@nail-gad:~/Desktop/HW_20$ sudo rmmod test1
gadirov@nail-gad:~/Desktop/HW_20$ lsmod
Module                Size  Used by
exfat                  110592  1
nfnetlink_queue       28672  1
nft_queue              12288  2
xt_conntrack           12288  3
xt_MASQUERADE          16384  3
bridge                425984  0
stp                   12288  1 bridge
```

```
gadirov@nail-gad:~/Desktop/HW_20$ sudo dmesg | tail -n1
[sudo] пароль для gadirov:
[61868.649565] My test module unloaded!!!!
```

4) создаем модуль с файлами устройств

```
#include <linux/module.h>
#include <linux/kernel.h>
#include <linux/fs.h>
#include <linux/rwlock.h>
#include <linux/uaccess.h>
// #include <linux/proc_fs.h>
// #include <linux/sysfs.h>
// #include <linux/string.h>
// #include <linux/kobject.h>

static int major = 0;
static rwlock_t lock;
static char test_string[15] = "Hello!\n\n";

static ssize_t test_read(struct file *fd, char __user *buff, size_t size, loff_t *off)
{
    size_t rc;

    read_lock(&lock);
    rc = simple_read_from_buffer(buff, size, off, test_string, 15);
    read_unlock(&lock);

    return rc;
}

static ssize_t test_write(struct file *fd, const char __user *buff, size_t size, loff_t *off)
{
    size_t rc = 0;
    if (size > 15)
        return -EINVAL;

    write_lock(&lock);
    rc = simple_write_to_buffer(test_string, 15, off, buff, size);
    write_unlock(&lock);

    return rc;
}

static struct file_operations fops = {
    .owner = THIS_MODULE,
    .read = test_read,
    .write = test_write
};

int init_module(void)
{
    pr_info("Test module is loaded.\n");
    rwlock_init(&lock);
    major = register_chrdev(major, "test2", &fops);

    if (major < 0)
        return major;
    pr_info("Major number is %d.\n", major);

    return 0;
}

void cleanup_module(void)
{
    unregister_chrdev(major, "test3");
    pr_info("Test module is unloaded.\n");
}

MODULE_LICENSE("GPL");
```

```
gadirov@nail-gad: ~/Desktop/HW_20$ make
make -C /lib/modules/6.17.0-35-generic/build M=/home/gadirov/Desktop/HW_20 modules
make[1]: вход в каталог «/usr/src/linux-headers-6.17.0-35-generic»
make[2]: вход в каталог «/home/gadirov/Desktop/HW_20»
warning: the compiler differs from the one used to build the kernel
The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
You are using: gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
CC [M] test2.o
MODPOST Module.symvers
WARNING: modpost: missing MODULE_DESCRIPTION() in test2.o
CC [M] test2.mod.o
LD [M] test2.ko
BTF [M] test2.ko
Skipping BTF generation for test2.ko due to unavailability of vmlinux
make[2]: выход из каталога «/home/gadirov/Desktop/HW_20»
make[1]: выход из каталога «/usr/src/linux-headers-6.17.0-35-generic»
gadirov@nail-gad: ~/Desktop/HW_20$
```

5) смотрим на номер для записи в систему

```
gadirov@nail-gad:~/Desktop/HW_20$ sudo insmod test2.ko
gadirov@nail-gad:~/Desktop/HW_20$ sudo dmesg | tail -n2
[62959.224360] Test module is loaded.
[62959.224364] Major number is 503.
gadirov@nail-gad:~/Desktop/HW_20$
```

6) результат

```
gadirov@nail-gad:/dev$ sudo rm test3
gadirov@nail-gad:/dev$ sudo mknod test3 c 503 0
gadirov@nail-gad:/dev$ sudo chmod 666 test3
gadirov@nail-gad:/dev$ cat test3
Hello!

gadirov@nail-gad:/dev$
```

```
gadirov@nail-gad:/dev$ cat test3
Hello!

gadirov@nail-gad:/dev$ sudo echo "dd" > test3
gadirov@nail-gad:/dev$ cat test3
dd

gadirov@nail-gad:/dev$
```